25901546 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSTA201 - STATISTICAL INFERENCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. Give Chebychev's inequality for finding upper bound of a probability . [K] / [CO1]

2. Let X_1 and X_2 be two independent observations from normal population with mean μ and variance σ2, if $t_1 = 2X_1 - X_2$ and $t_2 = 7X_1 - 6X_2$, find the relative efficiency of t_1 with respect to t_2. [A] / [CO2]

3. Define Unbiasedness of an estimator. [K] / [CO2]

4. A random sample of size n is taken from a normal population with mean 0 and variance σ^2.What is the minimum variance unbiased estimator of σ2 and what is its minimum variance ? [A] / [CO2]

5. Find the maximum likelihood estimator for the parameter θ , if $f(x) = \frac{1}{\theta} \quad 0 \leq x \leq \theta$. [A] / [CO2]

6. Define the likelihood ratio in the context of the Neyman–Pearson most powerful test procedure. [U] / [CO3]

7. How do you calculate the expected frequencies in a chi-square test for independence? [K] / [CO5]

8. Explain one sample sign test. [U] / [CO5]

9. State the test statistic for testing the population proportion. [K] / [CO4]

10. What is the test statistic used in paired t-test? [K] / [CO4]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. If X_i is a random variable which assumes values i and -i with equal probabilities, show that the law of large numbers cannot be applied to sequence $X_1, X_2, ..., X_n$. [A] / [CO1]

12. Derive the confidence interval for the population mean μ of a normal population $N(\mu, \sigma^2)$, when the population standard deviation σ is known. [U] / [CO2]

13. Explain Fisher-Neyman factorization criterion. [U] / [CO2]

14. Explain the procedure to conduct Kruskal-Wallis test. [An] / [CO5]

15. Two independent observations X_1 and X_2 are made on a random variable X with density function $f(x;\theta) = \frac{1}{\theta}e^{-\frac{x}{\theta}}, \quad 0 \leq x < \infty,\ \theta > 0$. To test the null hypothesis $H_0 : \theta = 2$ against $H_1 : \theta = 4$, it is decided to accept H_0 if $X_1 + X_2 \leq 9.5$ and reject otherwise. Obtain the level of significance and power of the test. [A] / [CO3]

16. A sample of 15 observations is given: 12.1, 11.8, 12.5, 12.3, 11.9, 12.4, 12.2, 12.0, 12.6, 11.7, 12.3, 12.1, 12.5, 12.2, 12.0.Apply the Anderson–Darling test to check whether the data can be assumed to come from a normal distribution. [An] / [CO5]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. a) Explain the maximum likelihood method of estimation.What are the properties of maximum likelihood estimators? [U] / [CO2]

    b) Explain the moment method of estimation. What is the advantage of moment method of estimation?

18. Explain the test procedure of one-way ANOVA. [U] / [CO4]

[12x1 = 12]